Vedansh Somani

Undergraduate Student

vedansh.somani.study@gmail.com | +91 9583221281 | Bengaluru, India

linkedin.com/in/vedansh-somani | github.com/sudoVed | Portfolio - vhades.dpdns.org

Profile

Robotics & AI undergraduate specializing in machine learning, reinforcement learning, and autonomous systems. Experienced in developing deep learning models, RL agents, and ROS2-based robotic systems, with additional experience building full-stack applications.

Education

BE - Robotics and AI | CGPA: 9.11 — 09/2023 – Present
Dayananda Sagar College of Engineering — Bengaluru, India

Professional Experience

STEM Trainer — 04/2024 – 04/2025
Beylabee — Bengaluru, India

- Delivered hands-on STEM and programming workshops, guiding students through robotics, electronics, and software-based projects.
- Designed and explained technical concepts through practical demonstrations, improving student understanding of engineering and computational principles.

Skills

Machine Learning and Reinforcement Learning
- PyTorch
- PPO
- DQN (Deep Q-Learning)
- Stable-Baselines3
- Neural Networks
- Deep Learning
- Reward Shaping
- Curriculum Learning
- CNN
- Model Quantization

Robotics and Autonomous Systems
- ROS2
- Nav2
- SLAM
- Motion Planning
- Robot Control
- MuJoCo

Programming Languages
- Python
- C++
- HTML/CSS
- JavaScript
- Nodejs
- React
- Flutter

Embedded Systems and Tools
- ESP32
- Arduino
- Microcontrollers
- Git
- Linux
- TensorBoard

Projects

STRIDE v1 - RL Based Quadruped Locomotion (Unitree Go2) 05/2026 – 05/2026

Python, MuJoCo, PPO, PyTorch

- Developed a PPO-based locomotion framework for a 15 kg Unitree Go2 quadruped in MuJoCo, training a single neural policy from scratch for gait generation and disturbance recovery using PMTG-based gait generation, residual joint control, and PD tracking.
- Evaluated robustness against impulsive mid-gait disturbances across 8 directions, achieving 100% recovery at 90 N (~0.6× body weight) on flat terrain and 100%/91.7% recovery at 75/90 N on procedurally generated uneven terrain (0.15 m elevation variation).

Chess Engine — Browser NNUE Engine 06/2026 – 06/2026

C++, WebAssembly, SIMD, PyTorch

- Built a chess engine from scratch in C++ (bitboard move generation, alpha-beta search with transposition tables, quiescence, null-move pruning, and late-move reductions), compiled to WebAssembly to run backend-free in the browser.
- Trained a HalfKP NNUE evaluator in PyTorch and optimized inference with incremental accumulator updates, integer quantization, and WASM SIMD, cutting per-evaluation cost ~3–4× (depth-10 search in ~330 ms), verified bit-exact against full rebuilds.

Chain Reaction + RL 04/2026 – 04/2026

Python, PyTorch, Deep Q-Learning

- Recreated the Chain Reaction board game in Python/Pygame and trained a DQN agent with a CNN-based state representation, curriculum learning, reward shaping, and heuristic-guided action filtering.
- Achieved a 95% win rate vs. random, 81.5% vs. defensive, and 57% vs. greedy opponents over ~10,000 training episodes, with a self-simulation risk filter that evaluates the opponent's best reply before each move.

Autonomous Mobile Robot 02/2026 – 05/2026

ROS2, Nav2, SLAM, LiDAR

- Developed an autonomous mobile robot using ROS2, LiDAR-based SLAM, and Nav2 for localization and navigation.
- Integrated SLAM, localization, path planning, and obstacle avoidance pipelines on a differential-drive platform.

Chat Application 12/2025 – 02/2026

Flutter, Node.js, Socket Programming

Built a full-stack mobile chat app with real-time WebSocket messaging, persistent database storage, and encrypted authentication.

Courses

Neural Networks - Zero to Hero 06/2026 – Present

Andrej Karpathy Bengaluru

Implemented neural networks from scratch, covering backpropagation, optimization, MLPs, CNNs, and transformer architectures using PyTorch.

Data Analysis with Python 04/2026 – 04/2026

IBM Developer Skills Network Bengaluru

Covered data preprocessing, exploratory data analysis, statistical analysis, and visualization using Python libraries including NumPy, Pandas, and Matplotlib.